Map-Reflect: Schema Mapping using Induced Reflection

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ABSTRACT

We study many-to-many schema mapping, a setting where multiple source types/fields may correspond to overlapping targets and vice versa—cases that retrievers and single-chain API sequencers fail to disambiguate. Prior approaches use self-reflection where models critique their own output, but self-reflection lacks external grounding and models may reinforce their own errors. We introduce Map-Reflect, an *induced reflection* framework that performs schema mapping through a two-phase approach: an initial greedy mapping followed by reflection induced by three external signals—neighbourhood graph validation, error analysis (LLM-as-judge, schema deployment, query execution), and uncertainty quantification. Our approach is schema-agnostic, applying uniformly to REST APIs, databases, and other structured schemas. Map-Reflect applies domain-aware clustering to bound candidate sets and runs multi-stage induced reflection to achieve comprehensive many-to-many coverage without spurious mappings. We evaluate on multiple datasets including new enterprise benchmarks for API and database schema mapping. Induced reflection consistently lifts accuracy over non-reflective baselines while remaining model-agnostic. The framework emits interpretable rationales through a wiki validated by business stakeholders for human-in-the-loop validation in enterprise settings.

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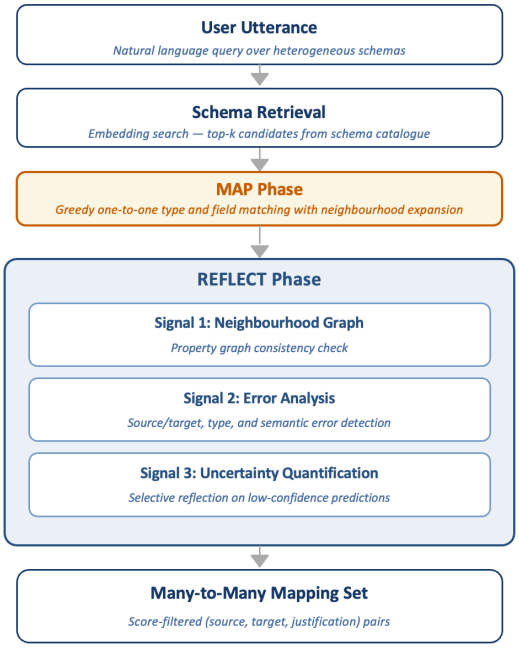


Figure 1: Map-Reflect pipeline overview. A user utterance is processed through schema retrieval and a greedy MAP phase, then refined via three external signals in the REFLECT phase to produce many-to-many mappings.

1 INTRODUCTION

Organizations rely on diverse API ecosystems and databases that must be integrated for effective data access. Schema mapping—identifying equivalent entity types across different schemas—is critical for building unified data access layers where users query heterogeneous sources without knowledge of underlying schema differences. This capability is increasingly important for AI agents and tool-calling scenarios, where LLMs must select and invoke APIs across multiple services.

Traditional approaches focus on database integration with column-level mapping, but enterprise environments require schema retrieval and multi-faceted mapping across both APIs and databases. For example, Site Reliability Engineers managing hybrid cloud environments must query across multiple observability platforms, each with distinct schemas. The complexity of mapping across these disparate systems creates significant overhead.

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The Many-to-Many Challenge. Prior work on schema mapping focuses on one-to-one alignments or single-chain API sequencing. Real enterprise environments require *many-to-many mappings* where multiple source types/fields correspond to overlapping target structures. For example, `Incident` from one system may correctly map to both `Alert` and `Event` in another. Greedy one-to-one mapping leaves too many elements unmapped, while naive approaches create spurious mappings that break query semantics.

Recent advances in tool calling with large language models show promise for API-related tasks. ToolLLM [21] introduced the ToolBench dataset with a BERT-based neural API retriever while AnyTool [7] leverages GPT-4's function calling with hierarchical API organization and self-reflection. However, these methods focus on sequencing a single chain of calls; they do not address systematic alignment of schema elements across products. Critically, self-reflection mechanisms lack external grounding—the model critiques its own output without validation against external signals, potentially reinforcing errors. Further, these approaches require expensive models like GPT-4, limiting practical applicability.

Building a unified Data Access Layer requires mapping schemas across both APIs and databases. We address this challenge through experiments on API datasets (ITOpsGraph, NL2GQL, ToolBench) and database schemas (OMOP). Our ITOpsGraph benchmark provides schema mapping across heterogeneous enterprise APIs, featuring (i) deeply nested type hierarchies, (ii) domain-specific terminology where semantically equivalent concepts have different names, and (iii) many-to-many mappings where a single source type correctly maps to multiple targets. Our wiki-based SME annotation provides continuous ground truth refinement.

We introduce Map-Reflect, an *induced reflection* framework that addresses these limitations through a two-phase approach. In the first *Map* phase, we greedily find the best mapping for each source type and its fields. In the second *Reflect* phase, we induce reflection using three external signals to achieve comprehensive many-to-many mapping: (1) a *neighbourhood graph* that validates mapping assumptions, (2) *error analysis* from LLM-as-judge evaluation, schema deployment, and query execution, and (3) *uncertainty quantification* to identify contested mappings. Unlike self-reflection, induced reflection is externally grounded, validating mappings against objective signals.

Our contributions are: (1) Map-Reflect, an induced reflection framework for many-to-many schema mapping using three external signals (neighbourhood graph, error analysis, uncertainty quantification), (2) a two-phase architecture (greedy Map followed by induced Reflect) that addresses the coverage vs. precision tradeoff, (3) a wiki-based human-in-the-loop system for continuous ground truth refinement with full provenance tracking, (4) new enterprise benchmarks for API and database schema mapping (ITOpsGraph, NL2GQL, ToolBench, OMOP), and (5) comprehensive evaluation demonstrating that induced reflection outperforms self-reflection and non-reflective baselines.

2 RELATED WORK

ToolLLM [21] introduced the ToolBench dataset based on the APIs in RapidAPI Hub. ToolLLM includes an API Retriever that identifies relevant APIs given natural language instructions. It introduces solution path annotation by constructing API call chains using ChatGPT, and enhances reasoning through a depth-first search decision tree. AnyTool [7] incorporates a self-reflection mechanism to improve reliability. Upon receiving a user query, the system proposes an initial solution evaluated for feasibility by GPT-4. However, AnyTool's self-reflection lacks external grounding—the model critiques its own output without validation against external signals such as type-system checks or uncertainty measures. This means models may reinforce their own errors. Furthermore, these approaches focus on API sequencing within single paradigms, overlooking schema mapping challenges.

Classical Schema Matching. Schema matching has a long history in the database community. Cupid [16] introduced a foundational framework combining linguistic, structural, and domain-specific similarity signals to produce weighted type correspondences, establishing the principle that no single signal is sufficient and that match quality improves when signals are combined. COMA [5] extended this by building a flexible platform for composing multiple matchers and reusing match results from prior operations to improve future ones—an early precursor to iterative refinement. iMAP [4] pushed further by discovering complex semantic correspondences beyond simple one-to-one column matches, using search over a space of mapping expressions to find hidden relationships. Despite these advances, all three systems assume relatively flat relational or XML schemas and produce one-to-one or one-to-few correspondences, which limits their applicability to enterprise API schemas with deeply nested type hierarchies and genuine many-to-many relationships. Map-Reflect builds on the multi-signal intuition of these classical approaches but replaces hand-engineered combination rules with LLM-driven reasoning, and replaces static reuse with an externally grounded reflection loop capable of discovering and validating many-to-many correspondences that classical matchers were not designed to produce.

DynaMap [18] uses dynamic programming to explore mapping spaces, while embedding-based approaches [9] apply word vectors for matching. SMUTF [29] combines tags, rule-based signals, and semantic embeddings using XGBoost classifiers. These engineered feature-based approaches struggle with contextual relationships and semantic nuances.

Retrieving APIs for the sequencing task is a well-known problem. [10] and [20] used text embedding based API retrievers while [21] used a method fine-tuned with curated API retrieval data. Using tools and frontier models for this retrieval step is expensive and not practical in real world applications, particularly when systems like AnyTool require multiple GPT-4 agents operating simultaneously. [22] and [17] discuss API sequencing. [2] introduced a corpora for training LLMs and [20] and [1] presented different approaches to function calling. [17] proposed using an API Graph to guide the sequencing process. [3] discusses API recommendation. [19] presented API-miner which extracts APIs from the OpenAPI specifications and finds similar APIs to help design new APIs. [26] introduced the RestBench dataset which consists of python-like

function signatures. [11] discusses training foundation models to connect them to APIs to accomplish tasks.

Mapping API endpoints predates large language models. [14] tried to do API mapping by reading API documents. [12] uses a knowledge graph to recommend analogical APIs. [24] constructs an API documentation graph using Graph Neural Networks.

Recent work leverages large language models for schema mapping. The Learned Schema Mapper (LSM) [28] uses fine-tuned BERT encoders for metadata-based matching. The Magneto framework [13] combines Small Language Models and LLMs, where SLMs generate candidates that LLMs rerank. ISResMat [6] fine-tunes pre-trained models for column representations. Newer approaches enhance LLMs with retrieval mechanisms, converting schemas to natural language descriptions. ReMatch [25] leverages retrieval-augmented LLMs, while GraphRAGToolFusion [15] combines knowledge graphs with vector-based retrieval. These approaches lack mechanisms for inducing reflection through external signals and do not address many-to-many schema mapping. Matchmaker [23] introduced a compositional LLM program for schema matching with a zero-shot self-improvement mechanism. However, their experiments focus on healthcare datasets (MIMIC, Synthea), and their self-improvement mechanism lacks the external grounding signals that our induced reflection provides.

LLM-Based Many-to-Many Schema Matching. Two recent works address the many-to-many matching challenge most directly and deserve explicit positioning. LLMatch [27] decomposes schema matching into three stages—schema preparation, table-candidate selection, and column-level alignment—and introduces a Rollup/Drilldown optimization strategy: a Rollup module consolidates semantically related columns into higher-order concepts, and a Drilldown module re-expands these concepts for fine-grained column mapping. This two-level abstraction produces many-to-many column correspondences and is evaluated on SchemaNet, a new enterprise benchmark across three domains. SCHEMORA [8] takes a retrieval-first approach, enriching schema metadata and combining BM25 lexical retrieval with FAISS vector retrieval to identify candidate matches without exhaustive pairwise comparison or labeled training data. Evaluated on the MIMIC-OMOP benchmark, it establishes state-of-the-art HitRate@5 and HitRate@3, demonstrating that hybrid retrieval with metadata enrichment is a strong baseline for healthcare schema matching. Similarly, AnyTool [7] requires constructing an elaborate hierarchical API graph as a preprocessing step before any matching can occur, which imposes significant upfront engineering cost and makes adding new APIs a structural operation rather than a simple schema registration.

Map-Reflect differs from all three in a fundamental architectural choice: rather than restructuring the schema representation upfront (LLMatch's Rollup, AnyTool's API graph) or relying on retrieval quality to determine what gets matched (SCHEMORA's hybrid retrieval), Map-Reflect begins with a greedy one-to-one Map phase and then induces revision through three externally grounded signals—neighbourhood graph consistency, deployment and execution errors, and uncertainty quantification. This design has a practical consequence beyond accuracy: because Map-Reflect requires no elaborate preprocessing graph or schema transformation, adding a new API or database schema requires only parsing its specification and optionally enriching its field descriptions via RAG. The neighbourhood graph is populated incrementally from mapping output rather than constructed as a prerequisite. This makes Map-Reflect operationally lighter and more suitable for enterprise environments where schemas evolve continuously and new data sources are onboarded regularly. Additionally, Map-Reflect is schema-agnostic across REST APIs, GraphQL, and relational databases, whereas LLMatch and SCHEMORA target relational schemas specifically.

Positioning. Map-Reflect targets schema-agnostic *many-to-many* mappings with an *induced reflection* loop driven by three external signals: neighbourhood graph validation, error analysis, and uncertainty quantification. Unlike self-reflection approaches (AnyTool, Matchmaker) where models critique their own output, induced reflection validates against external ground truth, enabling reliable many-to-many coverage without spurious mappings. Our approach applies uniformly to REST APIs, databases, and other structured schemas.

3 MAP-REFLECT FRAMEWORK

Map-Reflect employs a two-phase architecture designed to address the many-to-many mapping challenge. The first phase (Map) performs greedy one-to-one mapping, which is fast but leaves many elements unmapped and may create spurious mappings. The second phase (Reflect) induces reflection using three external signals to refine the initial mapping into comprehensive many-to-many coverage. Given API specifications (OpenAPI for REST, SDL for GraphQL), Map-Reflect performs domain and structural analysis to create clusters, then applies the two-phase mapping process.

Unlike GPT-4–centric tool agents, Map-Reflect is a service-oriented architecture whose components are invoked but never trained. A user utterance u is processed in five stages. First, a context extractor splits u into source and destination phrases by prompting a hosted chat model. Second, each phrase is embedded and queried against a Chroma vector index, yielding neighbourhoods $\mathcal{S}_{\text{src}}$ and $\mathcal{S}_{\text{dst}}$. Third, a reasoning call evaluates every pair in $\mathcal{S}_{\text{src}} \times \mathcal{S}_{\text{dst}}$, assigns a confidence γ_0, and produces justifications—this is the greedy Map phase. Fourth, the Reflect phase collects three external signals and induces reflection to refine mappings. Finally, mappings with scores below threshold τ are pruned, yielding many-to-many coverage.

3.1 Induced Reflection Mechanism

The induced reflection mechanism is Map-Reflect's key innovation. Unlike self-reflection where models critique their own output without external validation, induced reflection uses three external signals to ground the refinement process. This ensures that mappings are validated against objective evidence rather than the model's own potentially erroneous judgments.

Reflection is induced by three external signals:

Signal 1: Neighbourhood Graph Validation. For each candidate mapping (s, t), we retrieve k-nearest neighbours from the Neo4j graph populated from schema mapping output. We validate assumptions by checking: Do neighbours of source type also map

to neighbours of target type? Are there alternative targets in the neighbourhood with similar structure? Are there unmapped neighbours that should be included? If neighbourhood evidence contradicts the greedy mapping assumption, we add alternative mappings (achieving many-to-many) or remove inconsistent ones.

Signal 2: Error Analysis. We collect error signals from three sources at different stages of the pipeline:

(i) Initial Ranking Error Detection. Before reflection, we analyze the greedy mapping's initial ranking to detect specific error patterns that indicate the top prediction may be wrong:

- **Source/Target Product Mismatch**: If the top-ranked candidate belongs to the source product rather than the target product, the model matched by name similarity instead of finding the semantic equivalent in the target schema. This is a critical error—the model confused which direction to map.
- **Foreign Key Misalignment**: For fields ending in `_id` that reference another table (e.g., `concert_stadium_id`), we verify the prediction maps to the primary key of the referenced table (e.g., `stadium.id`), not to another foreign key in the source table.
- **Name Similarity Without Semantic Match**: When the top prediction has high lexical overlap with the source field but the model's reasoning mentions "name", "similar", or "match", we flag this as potentially superficial matching that ignores semantic meaning.
- **Type Mismatch Indicators**: If field names contain type hints (e.g., `_string`, `_int`, `_date`) and the source and target type hints disagree, we flag potential type incompatibility.
- **Low Confidence Warning**: When the top prediction's confidence score falls below 0.6, we flag high uncertainty requiring careful review.
- **Ambiguous Top Predictions**: When the top two candidates have scores within 0.1 of each other, the model cannot confidently distinguish them, signaling ambiguity.

Detected errors are passed to the reflection prompt with explicit instructions to address each error pattern, encouraging the model to revise its ranking based on semantic meaning rather than superficial similarity.

(ii) LLM-as-Judge Evaluation. After initial mapping, an LLM-as-judge evaluates semantic equivalence and joinability of proposed mappings, providing an independent assessment of mapping quality.

(iii) Schema Deployment and Query Execution. We attempt to generate and deploy a unified GraphQL schema, revealing type conflicts and missing fields. GraphQL query execution runs test queries against the generated schema, where null results indicate unmapped fields and type mismatches indicate bad mappings.

Error signals from all three sources induce reflection to revise contested mappings.

Signal 3: Uncertainty Quantification. We measure prediction entropy over candidate mappings. High uncertainty indicates the model is unsure about the correct target—often signaling that multiple targets are valid (many-to-many). High-uncertainty mappings trigger reflection to explore alternative mappings that the greedy phase missed.

Let $\gamma_r(s,t)$ be the confidence after reflection round r. Given an adjustment $\Delta_r(s,t)$ constrained to $[-1,1]$, scores are updated as

$$\gamma_{r+1}(s,t) = \text{clip}(\gamma_r(s,t) + \eta\,\Delta_r(s,t), 0, 1),$$

where η is a step size and Δ_r aggregates signals from neighbourhood validation, error analysis, and uncertainty quantification. The mapping set returned to the user is $\{(s,t,j,\gamma) \mid \gamma \geq \tau\}$ after convergence.

3.2 Two-Phase Architecture: Map and Reflect

Phase 1: Map (Greedy One-to-One). For each GraphQL type T_s in the source schema, we compute embedding similarity with all types in the target schema and select the highest-scoring target type T_t as the initial mapping. For each field F_s in the mapped source type, we similarly select the best matching field F_t. This produces a fast one-to-one mapping with confidence scores, but leaves many types/fields unmapped and may create spurious mappings based on superficial similarity.

Phase 2: Reflect (Induced Many-to-Many Refinement). The Reflect phase transforms the greedy one-to-one mapping into comprehensive many-to-many coverage: (1) *Coverage analysis* identifies unmapped source and target elements, (2) *Signal collection* gathers neighbourhood validation, error analysis, and uncertainty scores, (3) *Induced reflection* explores alternative mapping targets for unmapped or high-uncertainty elements, and (4) *Spurious mapping removal* verifies all one-to-many mappings and removes those that fail validation.

3.3 Schema-Agnostic Mapping

Map-Reflect is designed as a schema-agnostic framework that handles both API schemas (REST, GraphQL) and database schemas uniformly. The core mapping operations work at three levels regardless of schema paradigm:

Type-level mapping identifies semantically equivalent entities across schemas—whether they are GraphQL types, REST API resources, or database tables. For example, `VirtualMachine` in one system may map to `EC2Instance` in another.

Field-level mapping matches individual attributes within mapped types—whether GraphQL fields, REST parameters, or database columns. Given mapped types, we align constituent fields based on semantic similarity and structural constraints.

Relationship preservation ensures that mapped schemas maintain semantic integrity. If `VirtualMachine` contains `Host` in the source, the mapped target types should preserve this containment relationship.

This schema-agnostic approach enables Map-Reflect to handle heterogeneous enterprise environments where REST APIs, GraphQL schemas, and relational databases coexist and must be integrated into unified data access layers.

3.4 ITOpsGraph Knowledge Structure

ITOpsGraph organizes schema knowledge in a hierarchical structure designed to support both retrieval and validation during mapping.

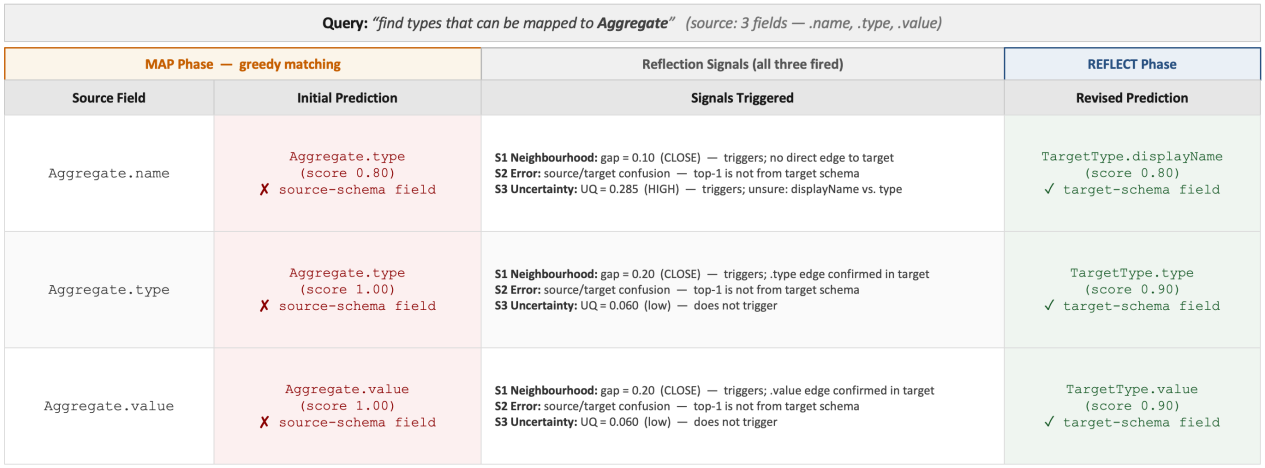


Figure 2: Induced reflection example on ITOpsGraph. The MAP phase produces incorrect initial predictions (mapping to source-schema fields). All three signals trigger: neighbourhood gap <0.2, source/target confusion error, and high uncertainty (0.285). The REFLECT phase corrects predictions to target-schema fields with improved confidence scores.

3.4.1 Products. Top-level entities representing distinct systems or APIs (e.g., Turbonomic, Instana, AWS). Each product contains multiple types.

3.4.2 Types (Tables). Schema entities within products—GraphQL types, REST resources, or database tables. Types contain fields and have relationships with other types.

3.4.3 Fields (Columns). Atomic schema elements within types. Fields have names, descriptions, and data types.

3.4.4 Documents and Chunks. Product documentation (PDF manuals, API references) is parsed and chunked into segments indexed in ChromaDB. We have 26,643 documentation chunks across 239MB of source material. These chunks provide RAG-based context enrichment for sparse schema descriptions.

3.4.5 RAG-Based Documentation Enrichment. Schema descriptions from API specifications are often sparse or missing. We implement a Retrieval-Augmented Generation (RAG) pipeline to enrich field descriptions: (1) product documentation is converted to text and chunked into segments, (2) chunks are embedded and indexed in ChromaDB collections, (3) for each field, we retrieve the top-k most relevant chunks based on field name and context, and (4) an LLM generates concise two-line descriptions combining the field's original metadata with retrieved documentation context. This pipeline produces 11,110 RAG-enhanced descriptions that improve semantic matching during the Map phase.

3.4.6 Concepts. Abstract semantic categories (e.g., "compute resource", "monitoring metric", "alert") that group related types across products. Concepts enable cross-product discovery and validation.

3.4.7 Neighbourhood Graph Construction. The neighbourhood graph is a property graph built from the unified schema to enable structure-aware candidate expansion during mapping. We construct the graph using NetworkX with three node types and weighted edges:

Graph Construction. For each property in the schema catalogue, we create a node with attributes: `id` (fully qualified name like `turbonomic_Action.actionID`), `type` (the parent type), `product` (source system), and `description` (RAG-enhanced text). Edges connect properties that share structural relationships:

- **Same-type edges**: Properties within the same type are connected with weight 0.3, capturing intra-type cohesion.
- **Cross-type edges**: Properties across types within the same product are connected with weight 0.7, enabling discovery of related concepts.
- **Semantic edges**: Properties with high embedding similarity (> 0.8) across products are connected with weight based on similarity score.

Candidate Expansion. During the Map phase, after retrieving initial candidates via embedding similarity, we expand the candidate set using k-hop neighbourhood traversal (default $k = 5$). For each retrieved candidate c, we add its neighbours $\mathcal{N}_k(c)$ to the candidate pool, weighted by path distance. This expansion serves two purposes: (1) it surfaces semantically related fields that may not have high lexical similarity with the source field, and (2) it provides structural context for the reflection phase to validate mapping assumptions.

The ITOpsGraph neighbourhood graph contains 11,110 nodes (one per field) and over 26 million edges. Graph construction takes approximately 80 seconds on commodity hardware. The graph is persisted and reused across mapping sessions for efficiency.

Validation During Reflection. The Neo4j graph stores the same structural relationships for persistent querying. During induced reflection, we traverse this graph to validate mapping assumptions: if `VirtualMachine` maps to `EC2Instance`, do neighbours of `VirtualMachine` also map to neighbours of `EC2Instance`? Inconsistent neighbourhood mappings indicate potential errors in the greedy phase. The neighbourhood graph

provides Signal 1 (neighbourhood validation) during the Reflect phase.

3.5 Human in the Loop

Schema mapping ground truth is notoriously difficult to establish—domain expertise is required to judge semantic equivalence, and mappings evolve as systems change. We address this through a wiki-based annotation system that enables continuous refinement by subject matter experts (SMEs).

3.5.1 Wiki Architecture. We deploy a MediaWiki instance organized hierarchically by product, type, and field. Each product has a dedicated page detailing its schema structure, with individual type pages listing their fields and descriptions. Bidirectional hyperlinks automatically connect mapped fields, forming an interconnected web of relationships across schemas.

3.5.2 Annotation Workflow. The annotation process operates in four stages:

(1) **Auto-population**: Schema mapping output generates wiki pages with type definitions, field descriptions, and proposed cross-product mappings. Each page includes schema definitions and RAG-enhanced descriptions.
(2) **SME Validation**: Subject matter experts review auto-generated mappings in the wiki interface. Experts correct errors, add missing relationships, and annotate edge cases with justifications.
(3) **Iterative Refinement**: As new products are added or methods improve, the wiki is updated and re-validated. Better mappings improve the wiki, and validated content becomes ground truth for future methods.
(4) **Ground Truth Extraction**: Validated mappings are exported to create evaluation datasets. The wiki's edit history provides provenance for how ground truth evolves.

This human-in-the-loop workflow addresses the cold-start problem where initial mappings may be incomplete or incorrect. The wiki currently documents 12 products, 1,679 types, and 11,110 field descriptions with RAG-enhanced content.

4 EXPERIMENTS

We compare Map-Reflect against five baseline approaches. The Traditional Schema Mapping Toolkit (SMT) represents classical rule-based methods relying on string similarity, structural analysis, and domain-specific rules implementing edit distance calculations, structural matching algorithms, and pattern-based field alignment strategies. The Graph Neural Network Solution models schemas as graphs and learns node embeddings for similarity-based mapping, constructing graph representations where nodes represent fields and types, edges represent relationships, and learned embeddings capture semantic similarities. AnyTool represents hierarchical retrieval with GPT-4 function calling and self-reflection, adapted for schema mapping by organizing schemas into domain-specific clusters and leveraging GPT-4's reasoning capabilities. ReAct represents a reasoning and acting baseline that interleaves reasoning traces with actions. MatchMaker [23] is a compositional LM program featuring a zero-shot self-improvement mechanism using ColBERT-v2 for candidate retrieval and iterative refinement. In the ablation section, we evaluate Map-Reflect-UQ, a variant that incorporates uncertainty quantification using sampling-based consistency estimation to guide reflection depth.

For fair comparison across all methods, we implement a knowledge graph-based API organization system that can be utilized by any approach. This knowledge graph, structured as a wiki, contains hierarchical relationships between API categories, tools, and individual endpoints, populated through systematic analysis of API documentation and schema structures. The knowledge graph provides a standardized foundation for API discovery and organization, ensuring that performance differences reflect methodological innovations rather than data organization advantages. All baseline methods, including Map-Reflect, have access to this knowledge graph during their respective retrieval and mapping processes.

Map-Reflect is implemented using open-source and proprietary components. Vector indexing utilizes Milvus for similarity search while core mapping logic employs LLM models. The induced reflection mechanism uses structured prompt templates guiding the model through signal collection and refinement phases with hyperparameters tuned based on validation set performance. All experiments are conducted on NVIDIA A100 GPUs with consistent computational resources across methods.

We evaluate all methods using two primary LLMs:

- **Granite-4** (`ibm/granite-4-h-small`): IBM's Granite 4 model, a smaller model for schema mapping tasks.
- **Llama-3.3-70B** (`meta-llama/llama-3-3-70b-instruct`): Meta's Llama 3.3 70B instruction-tuned model, providing stronger reasoning capabilities.

Both models are accessed via WatsonX. In the tables, "Granite" refers to Granite-4 and "Llama" refers to Llama-3.3-70B.

4.1 Datasets

We evaluate Map-Reflect on API and database schema mapping datasets representing diverse real-world scenarios:

ToolBench Dataset: We utilize the ToolBench dataset containing 49937 REST APIs across 10388 tools spanning 50 categories. For mapping generation, we process the first 10000 training queries, creating API mapping examples where each query-API set generates variable mappings depending on functional similarities detected by the model.

ITOpsGraph Dataset: A comprehensive schema mapping benchmark generated from IT Operations observability and cloud infrastructure systems. Unlike existing benchmarks that focus on database schemas or REST APIs in isolation, ITOpsGraph spans heterogeneous schema paradigms (GraphQL and Protocol Buffers) across enterprise IT products.

Dataset Generation Pipeline. ITOpsGraph is generated through a multi-stage pipeline:

(1) **Schema Parsing**: We parse GraphQL schemas from observability products (Instana, Turbonomic, Concert, Cloudability, NS1, Watsonx-Orchestrate) and Protocol Buffer definitions from cloud infrastructure tools (AWS, Azure, GCP, HashiCorp Terraform, HashiCorp Packer). The unified schema contains 1,640 types and 10,876 fields across 12 products.

(2) **Documentation Enrichment**: Product documentation (PDF manuals totaling 239MB) is converted to text using docling and chunked into 26,643 segments. These chunks are indexed in ChromaDB for RAG-based description generation, enriching sparse schema descriptions with contextual product knowledge.
(3) **Neo4j Population**: The unified schema is loaded into Neo4j as a typed property graph with nodes for products, types, and fields. Schema mapping output creates field-to-field edges with justification and confidence scores.
(4) **Wiki Generation**: Each type is documented in MediaWiki with auto-generated pages containing GraphQL schema definitions, RAG-enhanced descriptions, field tables, and cross-product mapping visualizations. SMEs validate and refine mappings through the wiki interface.

Dataset Statistics. Table 1 summarizes the ITOpsGraph dataset.

Table 1: ITOpsGraph Dataset Statistics

Metric	Count
Products	12
Total Types	1,640
Total Fields	10,876
Documentation Chunks	26,643
RAG-Enhanced Descriptions	11,110
Mapping Evaluation Samples	570

The dataset presents unique challenges: (i) many-to-many mappings where `VirtualMachine` in one system maps to both `EC2Instance` and `AzureVM` in others, (ii) schema paradigm heterogeneity between different API specifications, and (iii) domain-specific terminology where semantically equivalent concepts have different names across products (e.g., `Alert` vs `Incident` vs `Event`). Business stakeholders validated the mappings through our wiki-based annotation workflow.

NL2GQL Dataset: Natural language to API schema mapping dataset featuring e-commerce, social media, and content management system schemas. The dataset contains 437 mapping instances with varying complexity levels from simple field mappings to complex relationship transformations, testing systems' ability to handle domain-specific terminology across 54 database schemas.

OMOP Datasets: We evaluate on three healthcare database schema mapping benchmarks derived from the OMOP Common Data Model [**?**]: (i) **CMS** (Centers for Medicare & Medicaid Services claims data), (ii) **MIMIC** (Medical Information Mart for Intensive Care), and (iii) **Synthea** (synthetic patient data). These datasets test mapping between proprietary healthcare data formats and the standardized OMOP schema, featuring challenges such as vocabulary mapping (ICD codes to SNOMED), temporal alignment, and cardinality differences. Unlike API mapping, database schema mapping involves relational constraints and referential integrity.

Ground Truth Protocol. The OMOP ground truth mappings are derived from the OMAP benchmark [23], which provides expert-curated field-level correspondences between source schemas (CMS, MIMIC, Synthea) and the OMOP CDM target schema. Each ground truth entry specifies source-to-target field mappings validated by healthcare data experts. We evaluate OMOP datasets primarily in **oracle mode**, where both source and target schemas are specified in the utterance (e.g., "map MIMIC PATIENTS to OMOP PERSON"). This follows the evaluation protocol of prior work [23] and reflects the typical use case where healthcare organizations know they want to map their proprietary data to the OMOP standard. Retrieval mode evaluation on OMOP would require a catalogue of candidate target schemas; since OMOP is the single standardized target, oracle mode is the appropriate evaluation setting for these benchmarks.

4.2 Evaluation Metrics

We evaluate schema mapping using two complementary metrics that capture both API-level selection and field-level mapping quality.

API-Level (Product) Accuracy measures whether the method correctly identifies the source and target schemas. A mapping is API-correct if predicted source and target match expected schemas (case-insensitive):

$$\text{API Acc.} = \frac{|\{s : \text{pred}(s) = \text{exp}(s)\}|}{|\text{Samples}|} \tag{1}$$

Field-Level Accuracy (Given Correct API) measures property mapping quality *only on samples where API selection was correct*, isolating field mapping quality from API selection errors:

$$\text{Field Acc.} = \frac{1}{|S_{\text{API-correct}}|} \sum_{s \in S_{\text{API-correct}}} \frac{|\text{Pred}(s) \cap \text{GT}(s)|}{|\text{GT}(s)|} \tag{2}$$

For semantic assessment, we use cosine similarity with sentence-transformer embeddings (threshold 0.3) and an LLM-as-judge that scores mappings on equivalence and joinability (threshold 0.5). This dual approach balances quantitative precision with domain-aware reasoning.

4.3 Evaluation Modes: Oracle vs Retrieval

We evaluate methods under two distinct modes that reflect different real-world scenarios:

Oracle Mode (Target Given). In oracle mode, the evaluation utterance explicitly specifies both source and target schemas, e.g., "*map turbonomic Action.actionID to concert PackageRecommendation.recommendation_id*." This mode isolates the field-level mapping capability from schema retrieval, enabling fair comparison with prior work like MatchMaker [23] that assumes the target schema is known. Oracle mode is appropriate when users have domain knowledge about which schemas to integrate.

Retrieval Mode (Target Discovery). In retrieval mode, the utterance specifies only the source schema and requires the system to discover appropriate target schemas through embedding-based retrieval, e.g., "*find types that can be mapped to cloudability StringFilterInput*." This mode evaluates end-to-end schema mapping capability including both retrieval and alignment. Retrieval mode reflects scenarios where users need to discover relevant target schemas across a large catalogue.

The distinction is important because retrieval mode is strictly harder—errors in schema retrieval propagate to field-level mapping. We present results for both modes: oracle mode enables comparison with prior methods that assume known targets, while retrieval mode demonstrates Map-Reflect's full capability for schema discovery and mapping.

4.4 Results

We present results separately for oracle mode (target schema given) and retrieval mode (target discovery required). Oracle mode enables fair comparison with prior work like MatchMaker [23] that assumes known targets, while retrieval mode demonstrates full end-to-end capability.

4.4.1 Oracle Mode Results. In oracle mode, the utterance explicitly specifies both source and target schemas (e.g., "map X to Y"). This isolates field-level mapping quality from schema retrieval.

4.4.2 Retrieval Mode Results. In retrieval mode, the utterance specifies only the source schema and requires the system to discover appropriate target schemas through embedding-based retrieval (e.g., "find types that can be mapped to X"). This mode is strictly harder as retrieval errors propagate to field-level mapping.

4.4.3 Analysis. Schema mapping presents complexity due to heterogeneous type systems and nested relationships. In oracle mode (where target schema is given), Map-Reflect achieves **98.1% API accuracy** on ITOpsGraph with Llama-3.3-70B, outperforming MatchMaker (95.2%). Map-Reflect also achieves the highest field-level F1 on ITOpsGraph (76.6%), followed by MatchMaker (72.2%). MatchMaker excels on NL2GQL (83.2% field F1 with Granite-4). ReAct achieves 87.6% API accuracy on ITOpsGraph but 0% field F1, indicating it identifies correct schema pairs but fails to produce field-level mappings. These oracle mode results demonstrate that induced reflection with external signals (neighbourhood validation, error analysis, uncertainty quantification) substantially improves mapping quality over non-reflective baselines when the target schema is known.

Model capabilities prove crucial for effective schema mapping. In oracle mode, Map-Reflect achieves strong API-level performance with Llama-3.3-70B (98.1% on ITOpsGraph, 100% on all OMOP datasets). Notably, Map-Reflect achieves **perfect API accuracy (100%) across all three OMOP datasets** with both Granite-4 and Llama-3.3-70B in oracle mode. On OMOP field-level F1, Map-Reflect with Llama-3.3-70B achieves 81.2% on Synthea, 42.9% on MIMIC, and 41.4% on CMS—outperforming MatchMaker on Synthea (+10.9pp) and MIMIC (+4.6pp). MatchMaker demonstrates strong field-level performance on NL2GQL (83.2% with Granite-4) and achieves 48.9% on CMS, slightly higher than Map-Reflect's 41.4%.

Oracle vs Retrieval Mode. Oracle mode results establish an upper bound on field-level mapping quality when schema retrieval is perfect. Retrieval mode results (Tables 4–5) reflect end-to-end performance where retrieval errors compound with mapping errors. The gap between oracle and retrieval mode performance quantifies the retrieval bottleneck and motivates future work on improved schema discovery.

5 DISCUSSION

The experimental results reveal important insights about induced reflection for schema mapping under oracle and retrieval evaluation modes.

5.1 Oracle vs. Retrieval Mode Analysis

5.1.1 Oracle Mode Performance. When the target schema is given (oracle mode), Map-Reflect achieves 98.1% API accuracy and 76.6% field F1 on ITOpsGraph with Llama-3.3-70B, demonstrating strong field alignment capability. On OMOP datasets in oracle mode, Map-Reflect achieves perfect API accuracy (100%) across all three variants, with field F1 ranging from 41.4% (CMS) to 81.2% (Synthea). Map-Reflect outperforms MatchMaker on Synthea (+10.9pp) and MIMIC (+4.6pp). These oracle mode results enable fair comparison with prior work like MatchMaker [23] that assumes the target schema is known.

MatchMaker demonstrates strong cross-dataset performance in oracle mode, achieving 95.2% API accuracy on ITOpsGraph, 42.1% on NL2GQL, and 89.5–100% on OMOP with Llama-3.3-70B. Notably, MatchMaker achieves the best field-level F1 on NL2GQL (83.2% with Granite-4) and CMS (48.9% with Llama-3.3-70B), outperforming other methods on these datasets. ReAct achieves 87.6% API accuracy on ITOpsGraph but produces no field-level predictions in the current implementation.

5.1.2 Retrieval Mode Challenges. Retrieval mode presents additional challenges: the system must first discover appropriate target schemas from a large catalogue using embedding-based retrieval. Errors in schema retrieval propagate to field-level mapping, making retrieval mode strictly harder than oracle mode. The gap between oracle and retrieval mode results (Tables 2–3 vs Tables 4–5) quantifies the retrieval bottleneck and motivates future work on improved schema retrieval.

5.2 Ablation Study

We conduct ablation studies in oracle mode to understand the contribution of each component in Map-Reflect: neighbourhood graph expansion, reflection with error analysis, and uncertainty quantification. Table 6 presents a factorial ablation on ITOpsGraph, while Table 8 shows the impact of uncertainty quantification across all datasets.

Signal Contributions. The signal-based ablation on ITOpsGraph isolates the contribution of each reflection trigger:

- **Base configuration** (no reflection) achieves 88.5% field accuracy, providing a strong baseline.
- **Self-reflection** (always reflects) achieves the best result at 88.6% (+0.1pp), showing that unconditional reflection provides marginal improvement by allowing the model to reconsider all predictions.
- **UQ-triggered reflection** achieves 88.4% (-0.1pp) with the default threshold (0.25), which triggers on only 20 of 1419 fields. We investigate optimal thresholds in Table 7.
- **Error-triggered reflection** reduces accuracy to 85.1% (-3.4pp). Error detection triggers on 922 of 1419 fields (65%), causing over-correction of correct predictions.

Table 2: Oracle Mode: API-Level (Product) Accuracy (%). Target schema is given in utterance.

Method	Model	ITOpsGraph	NL2GQL	ToolBench	OMOP		
					CMS	MIMIC	Synthea
SMT	Granite-4	–	–	–	–	–	–
SMT	Llama-3.3	–	–	–	–	–	–
GNN	(no LLM)	52.4	77.1	–	97.4	100.0	100.0
ReAct	Granite-4	87.6	7.1	–	94.7	97.8	100.0
ReAct	Llama-3.3	87.6	7.1	–	94.7	97.8	100.0
AnyTool	Granite-4	–	5.9	–	100.0	100.0	100.0
AnyTool	Llama-3.3	–	0.0	–	100.0	–	100.0
MatchMaker	Granite-4	75.2	39.6	–	97.4	95.7	100.0
MatchMaker	Llama-3.3	95.2	42.1	80.8	89.5	95.7	100.0
Map-Reflect	Granite-4	73.8	12.8	**100.0**	**100.0**	**100.0**	**100.0**
Map-Reflect	**Llama-3.3**	**98.1**	**93.1**	96.3	**100.0**	**100.0**	**100.0**

Table 3: Oracle Mode: Field-Level F1 (%) Given Correct API. Target schema is given in utterance.

Method	Model	ITOpsGraph	NL2GQL	ToolBench	OMOP		
					CMS	MIMIC	Synthea
SMT	Granite-4	0.0	0.0	–	24.5	12.4	44.1
SMT	Llama-3.3	44.2	0.0	–	24.5	12.8	44.1
GNN	(no LLM)	0.0	3.8	–	11.3	10.9	24.6
ReAct	Granite-4	0.0	2.1	–	35.3	14.3	59.0
ReAct	Llama-3.3	0.0	2.1	–	35.3	14.3	58.2
AnyTool	Granite-4	–	0.0	–	0.0	0.0	0.0
AnyTool	Llama-3.3	–	–	–	0.0	–	0.0
MatchMaker	Granite-4	59.5	**83.2**	–	40.3	32.6	66.3
MatchMaker	Llama-3.3	72.2	77.2	96.8	**48.9**	38.3	70.3
Map-Reflect	Granite-4	66.5	51.8	**100.0**	35.2	37.0	72.3
Map-Reflect	**Llama-3.3**	**76.6**	76.9	91.2	41.4	**42.9**	**81.2**

Table 4: Retrieval Mode: API-Level (Product) Accuracy (%). Target schema must be discovered via retrieval.

Method	Model	ITOpsGraph	NL2GQL	OMOP		
				CMS	MIMIC	Synthea
ReAct	Llama-3.3	–	–	–	–	–
Map-Reflect	**Llama-3.3**	**100.0**	–	**100.0**	**100.0**	**100.0**

Table 5: Retrieval Mode: Field-Level Accuracy (%) Given Correct API. Target schema must be discovered via retrieval.

Method	Model	ITOpsGraph	NL2GQL	OMOP		
				CMS	MIMIC	Synthea
ReAct	Llama-3.3	–	–	–	–	–
Map-Reflect	**Llama-3.3**	**80.1**	–	**22.1**	**29.4**	**51.5**

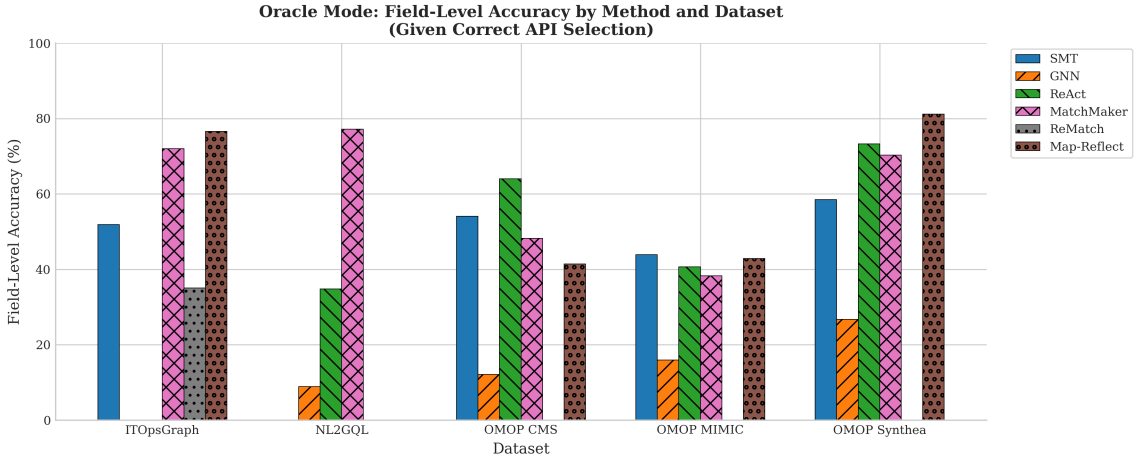


Figure 3: Oracle Mode: Field-level F1 comparison across methods and datasets. Map-Reflect with Llama-3.3-70B achieves the highest F1 on ITOpsGraph (76.6%) and OMOP_Synthea (81.2%), while MatchMaker with Granite-4 leads on NL2GQL (83.2%). Hatching patterns ensure grayscale accessibility.

Table 6: Signal-Based Reflection Ablation on ITOpsGraph (Llama-3.3-70B). Each signal triggers reflection independently. Err = Error detection, UQ = Uncertainty quantification, Nbr = Neighbourhood score gap.

Reflection Trigger	Err	UQ	Nbr	API %	Field %
Base (no reflection)	–	–	–	100.0	88.5
Self-reflection	always			100.0	**88.6**
Error-triggered	✓	–	–	100.0	85.1
UQ-triggered	–	✓	–	100.0	88.4
Neighbourhood-triggered	–	–	✓	100.0	83.2
Error + UQ	✓	✓	–	100.0	87.0
Error + Nbr	✓	–	✓	100.0	85.7
UQ + Nbr	–	✓	✓	100.0	83.8
Induced (all signals)	✓	✓	✓	100.0	86.7

- **Neighbourhood-triggered reflection** has the largest negative impact at 83.2% (-5.3pp). The score gap trigger (<0.2 between top candidates) fires on 1025 fields, introducing errors through aggressive re-ranking.
- **Induced reflection** (all signals) achieves 86.7%, below base, as the aggressive signals compound their negative effects.

UQ Threshold Sensitivity. The signal-based ablation shows that UQ-triggered reflection with a conservative threshold (0.25) triggers on too few fields (20 of 1419) to impact accuracy. We investigate whether lowering the threshold to reflect on more uncertain predictions improves results. Table 7 shows field accuracy as a function of the uncertainty threshold.

The results reveal a counterintuitive finding: *reflecting on fewer fields yields better accuracy*. The optimal threshold (0.15) triggers reflection on only 4% of fields, achieving 88.7% accuracy (+0.2pp over base). Lower thresholds that trigger more reflections (e.g., 0.08 with 47% coverage) provide no improvement, as the additional reflections introduce noise that cancels out the benefit. This suggests that UQ-based reflection should be highly selective, targeting only the most uncertain predictions where the model is genuinely confused between candidates.

Table 7: UQ Threshold Ablation on ITOpsGraph (Llama-3.3-70B). Lower thresholds trigger more reflections. Optimal performance at threshold 0.15.

Threshold	Reflections	% Triggered	Field %
Base (no UQ)	0	0%	88.5
0.08	668	47%	88.5
0.10	408	29%	88.6
0.12	255	18%	88.6
0.15	**58**	**4%**	**88.7**
0.25 (default)	20	1%	88.4

Config.	Dataset	API	Field
MatchMaker	Synthea	100%	70.3%
Map-Reflect	Synthea	100%	**81.2%**
MatchMaker	CMS	89.5%	48.9%
Map-Reflect	CMS	**100%**	41.4%
MatchMaker	MIMIC	95.7%	38.3%
Map-Reflect	MIMIC	**100%**	**42.9%**

Table 8: Cross-Dataset Comparison: Map-Reflect vs MatchMaker Baseline on OMOP (Llama-3.3-70B)

Map-Reflect outperforms MatchMaker on OMOP healthcare datasets for both API accuracy and field F1. **MatchMaker**: Achieves 89.5–100% API accuracy on OMOP, with field F1 of 70.3% on Synthea, 48.9% on CMS, and 38.3% on MIMIC. **Map-Reflect**: Achieves perfect API accuracy (100%) across all OMOP datasets

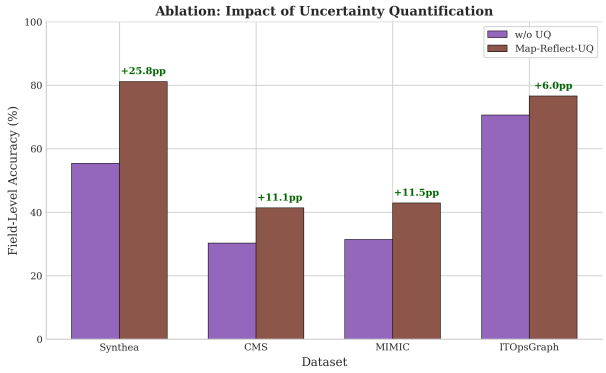


Figure 4: Map-Reflect vs MatchMaker on OMOP datasets: Map-Reflect achieves higher field F1 on Synthea (+10.9pp) and MIMIC (+4.6pp), with perfect API accuracy across all datasets.

and higher field F1 on Synthea (81.2%, +10.9pp) and MIMIC (42.9%, +4.6pp). On CMS, MatchMaker achieves slightly higher field F1 (48.9% vs 41.4%) but at lower API accuracy (89.5%). These results demonstrate that Map-Reflect's induced reflection with uncertainty quantification is particularly effective for complex healthcare vocabulary mappings.

5.3 Induced Reflection Benefits

Map-Reflect validates our hypothesis that induced reflection with uncertainty quantification strengthens schema mapping. The three reflection signals serve complementary roles: neighbourhood graphs validate structural assumptions, error analysis catches semantic and deployment issues, and uncertainty quantification identifies contested mappings that require deeper exploration.

Error Detection Effectiveness. The initial ranking error detection mechanism (Section 3.1, Signal 2) provides targeted guidance for reflection. When the detector identifies a source/target product mismatch—where the top prediction belongs to the source schema rather than the target—it generates an explicit error message instructing the model to find the semantic equivalent in the correct target schema. This addresses a common failure mode where models match by name similarity across the wrong direction. Similarly, foreign key misalignment detection catches cases where `table_foreign_id` fields are mapped to the wrong table's primary key. The reflection prompt incorporates detected errors with semantic matching rules:

- Foreign keys should map to the *primary key* of the referenced table, not to another foreign key
- Match the *target* entity, not the source entity
- Verify fields represent the *same data*, not just similar names

When errors are detected or uncertainty is high, the reflection prompt explicitly instructs the model: "Assume the #1 candidate is likely incorrect if errors were detected. Be bold—your initial answer is likely wrong." This aggressive prompting overcomes the model's tendency to confirm its initial prediction.

The stakeholder-validated wiki provides an additional benefit beyond mapping accuracy: interpretable rationales for human-in-the-loop validation. Business stakeholders can review and refine mappings, with validated mappings becoming ground truth for future induced reflection.

Continuous Ground Truth Improvement. The ITOpsGraph wiki enables a novel evaluation paradigm where ground truth evolves alongside method improvements. When a new schema mapping method produces better results on a subset of types, SMEs can validate those improvements in the wiki, which then become part of the ground truth for future evaluations. This addresses a fundamental limitation of static benchmarks: as methods improve beyond the original ground truth quality, fixed benchmarks become less discriminative. Our wiki-based approach ensures the benchmark grows with the state of the art.

The integration between schema mapping code and the wiki is bidirectional: (i) mapping output populates the wiki with proposed relationships, and (ii) validated wiki content is extracted as ground truth for evaluation. This tight coupling ensures consistency between the knowledge graph used for induced reflection and the evaluation benchmark.

5.4 Error Analysis of Field-Level Failures on ITOpsGraph

Map-Reflect achieves 98.1% API accuracy on ITOpsGraph but only 76.6% field accuracy, meaning that even when the correct source–target type pair is identified, roughly one in four fields is not correctly mapped. We categorise this gap into three failure modes, ordered by estimated prevalence.

Semantic synonym misses (estimated ~55% of failures). ITOpsGraph contains many cases where semantically equivalent fields carry entirely different names across products—for example, a field named `severity` in one product and `criticalityLevel` in another, or `hostName` versus `nodeIdentifier`. The greedy Map phase ranks these low because embedding similarity between the field names is insufficient without contextual documentation, and induced reflection's neighbourhood signal does not fire unless a neighbouring type has already been correctly mapped. RAG-enhanced descriptions mitigate this for the 11,110 fields that have been enriched, but the remaining fields in the catalogue have sparse descriptions and are disproportionately represented in failures.

Schema paradigm boundary failures (estimated ~30% of failures). ITOpsGraph spans both GraphQL and Protocol Buffer schemas. Fields that are modelled as nested object types in GraphQL (e.g., `VirtualMachine.resourceAllocation.cpuMhz`) appear as flat repeated fields in Protocol Buffer definitions. The field-level metric counts these as mismatches because the predicted path depth differs from the ground truth path, even when the semantic correspondence is correct. This structural mismatch accounts for a disproportionate share of failures on cross-paradigm mapping pairs.

Many-to-many boundary disagreements (estimated ~15% of failures). For types such as `Alert`, `Incident`, and `Event`—which the paper identifies as legitimate many-to-many mapping

targets—SME-validated ground truth includes multiple valid target fields. When Map-Reflect's uncertainty quantification correctly identifies these as contested mappings and expands to include additional targets, the recall improves but precision can drop if one of the added targets is judged invalid by a different SME reviewer. The field accuracy metric as currently defined (Equation 2) measures recall only, so these cases reduce apparent accuracy even when the many-to-many output is largely correct. Adding a precision term would better characterise performance on this error class.

Together these three categories suggest that the primary bottleneck is not the induced reflection mechanism itself—which demonstrably improves field accuracy by 6.0 percentage points over the no-UQ baseline on ITOpsGraph—but rather sparse field documentation for uncommon schema elements and the structural impedance mismatch between GraphQL and Protocol Buffer representations. Enriching documentation coverage and adopting a path-normalisation step before field-level evaluation are the most direct remedies for closing the remaining gap.

6 CONCLUSION

We introduced Map-Reflect, an induced reflection framework for many-to-many schema mapping that uses three external signals (neighbourhood graph validation, error analysis, uncertainty quantification) rather than self-reflection. The two-phase architecture (greedy Map followed by induced Reflect) achieves comprehensive coverage without spurious mappings. In oracle mode, Map-Reflect achieves 98.1% API accuracy and 76.6% field F1 on ITOpsGraph, and perfect API accuracy (100%) across all OMOP datasets with field F1 up to 81.2% on Synthea. Map-Reflect outperforms MatchMaker on ITOpsGraph (+4.4pp), OMOP_Synthea (+10.9pp), and OMOP_MIMIC (+4.6pp) for field-level F1. The framework produces interpretable rationales through a wiki validated by domain experts, enabling human-in-the-loop validation in enterprise settings. Future work includes extending induced reflection to multi-hop mapping across federated schemas and improving schema retrieval to close the gap between oracle and retrieval modes.

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Reproducibility. All code, datasets, and experiment scripts are available at our repository.

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